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**Prepared
by**

- 1) Muhammad Afiq Danish bin Mohd Hazni (A23CS0118)
- 2) Nabil Aflah Boo Binti Mohd Yosuf Boo Yong Chong (A23CS0252)
- 3) Najma Shakirah binti Shahrulzaman (A23CS0140)
- 4) Ling Yu Qian (A23CS0301)
- 5) Choh Jing Yi(A23CS0296)
- 6) Yasmin Batrisyia Binti Zahiruddin (A23CS0201)
- 7) Syarifah Dania Binti Syed Abu Bakar (A23CS0183)
- 8) Nur Firzana Binti Badrus Hisham (A23CS0156)

Section

01

Lecturer

Dr. Aryati binti Bakri

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Chapter 1: Introduction

1.1 Overview

Universiti Tun Hussein Onn Malaysia (UTHM) is a public university that provides various academic, administrative and support services to students. All these operations are made possible through a variety of information systems that are widely in use throughout the university mainly for the management of the teaching and learning processes, students' affairs, human resources, finance, and institutional management. As a large academic institution, UTHM operates in a complex business setting where applying the right tools and approaches to information systems and processes can readily result in extremely high productivity.

The goal of the project is to enhance the Enterprise Architecture (EA) of Universiti Tun Hussein Onn Malaysia (UTHM) so that there would be a great alignment of business processes and information systems throughout the university. The project aims to improve operational effectiveness and lay the framework for improved coordination across academic, administrative, and support systems by examining the current enterprise environment and using a detailed architectural approach.

1.2 Problem Statement

UTHM has successfully moved all the major academic, administrative, and support systems to the new platform, and the migration has gone over 70 percent so far. Still, the new university infrastructure of different faculties and departments has not yet been fully integrated and aligned with business processes, thus creating challenges in the integration process. The use of independent systems and the partial integration of different systems can bring on a variety of issues such as having data stored in multiple places, having inconsistent workflows, and being slower to deliver services.

In addition, UTHM's strategic ICT goals such as strengthening the infrastructure, providing integrated applications, and developing the workforce, in spite of their setting, are still not fully accessible due to the absence of a structured EA approach, it becomes quite challenging to make sure that all the systems are in total sync with the university's operational demands, strategic goals, and sustainability in the long run. Hence, the further development of Enterprise Architecture (EA) is required to reach system integration to the fullest, hassle-free business operations, and increased efficiency at UTHM.

1.3 Project Objectives

The main objectives of this project is to analyze, design and propose an enterprise-level system architecture for Universiti Tun Hussein Onn Malaysia (UTHM) by applying enterprise architecture principles and structured system design methodologies. This project aims to address the issues of managing multiple academic, administrative and support systems within a university environment by providing a proper architectural design. The objectives of this project are as follows:

1. To analyze the existing information systems used at UTHM that support key university operations such as academic management, teaching and learning, administrative processes and student services operations.
2. To design an enterprise architecture for UTHM using the TOGAF framework focusing on the alignment between business processes, data, applications and technologies.
3. To propose an integrated system design that improves data consistency, system interoperability and operational efficiency across UTHM's academic and administrative departments.
4. To provide a structured architectural reference that can support future system development and improvement initiatives at UTHM.

1.4 Project Scope

This project focuses on the analysis and design of an enterprise-level system architecture for Universiti Tun Hussein Onn Malaysia (UTHM) with emphasis on improving the alignment between business processes and information systems within the university environment. The scope of this project is carefully defined to ensure that the study remains focused on enterprise architecture design while remaining feasible within the given timeframe and academic requirements.

1.4.1 In-Scope

The scope of this project includes the following:

1. Analysis of UTHM's organizational structure and its core operational areas, particularly academic management, teaching and learning activities, administrative functions and student support services.
2. Identification of stakeholders and system users including students, academic staff, administrative staff and university management and analysis of how information systems support their daily operations.
3. Design of enterprise architecture based on the TOGAF framework following the Architecture Development Method (ADM).
4. Analysis of existing systems currently used at UTHM such as academic management systems, learning management systems, student assessment systems, staff related systems and student support systems.
5. Comparison between the existing system environment and the proposed enterprise architecture to highlight improvements in system integration, data management and operational efficiency.
6. Development of relevant system models and diagrams, including enterprise architecture diagrams, system architecture diagrams, use case diagrams and data models to support the proposed system design.

1.4.2 Out of Scope

The scope of this project excludes the following:

1. Full-scale system development, deployment or maintenance in a real operational environment.
2. Large-scale data migration, performance optimization, advanced system tuning.
3. Complete system coding for the whole enterprise systems beyond limited prototypes or demonstration purposes.
4. Security penetration testing, load testing and user acceptance testing.
5. Ongoing system monitoring, maintenance and post-deployment support activities.

1.5 Project Importance

The project importance are listed as follows:

1. To connect all disconnected university systems into one integrated architecture.
2. Help save time and resources by making daily operations faster and reducing duplicate work across departments.
3. Improves decision making by providing accurate, real-time data without any redundancy for better management decisions.
4. Meet the digitalization goals of the university.
5. Help create better user experience for students and staff to access services.

Chapter 2: Literature Review

2.1 Introduction

This chapter presents a detailed literature review related to the design and development of a University Enterprise Architecture (UEA) within the context of an enterprise-level information system. It commences by explaining the major concepts about the Enterprise Architecture (EA) which is the process of aligning the organizational processes, technology and information to facilitate the implementation of goals in a given institution. The chapter also reviews past research on EA applications in institutions of higher learning including the University of Lampung to learn about the most typical obstacles such as system fragmentation, data redundancy, insufficient to integrated academic or administrative systems. Then the chapter examines the chosen subsystems of the proposed university environment including student information, human resource, finance, subject registration, research management, learning management and analytics dashboards and points out how related subsystems have been used elsewhere. Lastly, it describes the technologies employed in the development of EA and these are the database solutions, development frameworks, enterprise systems and architecture standards like the TOGAF and the Zachman Framework. Altogether, this chapter establishes the theoretical and technical foundation that guides the design and implementation of the proposed enterprise system.

2.2 Enterprise Architecture (EA)

2.2.1 Definition and Concepts of EA

Enterprise Architecture (EA) is a comprehensive framework or blueprint that defines structure, processes, information and technology needed to effectively achieve an organization's business goals and objectives (Ardoq, 2024). In other words, EA is widely recognized as both a strategic discipline and a practice that enables organizations to align their business objectives, processes, information systems and technology infrastructure for long-term efficiency and adaptability. Instead of having a separate process between each department, EA helps shape a big-picture plan, to ensure every part of the business moves together toward its goals.

Most frameworks such as the Zachman Framework and TOGAF organize EA into four primary domains:

1. **Business Architecture:** Define the organization's strategy, governance, business processes and capabilities.
2. **Data Architecture:** Describe the logical and physical data assets, including their structure, management and flow throughout the organization.
3. **Application Architecture:** Encompasses applications, their interactions and their alignment with business processes.

- 4. Technology Architecture:** Details the technology infrastructure including hardware, networks and platforms that support applications, data and business processes.

EA provides both an “as-is”, meaning the current state and “to-be”, which means the future state of the organization (Tamm et al., 2011). This serves as the basis for organizational analysis and strategic transformation. Through this blueprint, EA identifies gaps, streamline processes and prioritizes investments and initiatives that support the desired target of enterprise evolution.

In simpler terms, EA is a management approach that helps an organization stay organized, aligned and adaptable. It brings together all parts of the organization from core processes, technology, people and strategies to ensure each component works smoothly with the rest. As the organization’s goals or environment change, EA serves as guidance for the evolution of the enterprise so the organization can remain competitive and effective.

2.2.2 Review of Existing Studies Related to University Enterprise Architecture

Previous studies on enterprise architecture in higher education have called attention to the need for more scalable and efficient subsystems to support complex academic and administrative services in the institution. Existing literature has shown that universities all over the world have transitioned from individual silo-based information systems to unified enterprise systems to improve overall operational efficiency and institutional decision making. Several studies have been conducted on different universities around the world as described in this section.

University of Lampung (Unila) is an Indonesian public university located in Lampung (Nama et al., 2017). The university has developed and designed a more flexible and modern IT infrastructure which to address the challenges in increasing management complexity, scattered and separated systems and was driven by the university's ambition to rank among Indonesia's top 10 universities by 2025 (Nama et al., 2017). This section will examine the methodology and approaches that Unila uses to achieve these strategic goals.

The study conducted at the University of Lampung identified Unila's core processes from several aspects which are education, research and community services. Under each of these aspects there are several processes such as Students Admission, Research Activity and Alumni Services, which altogether support the institutional operation.

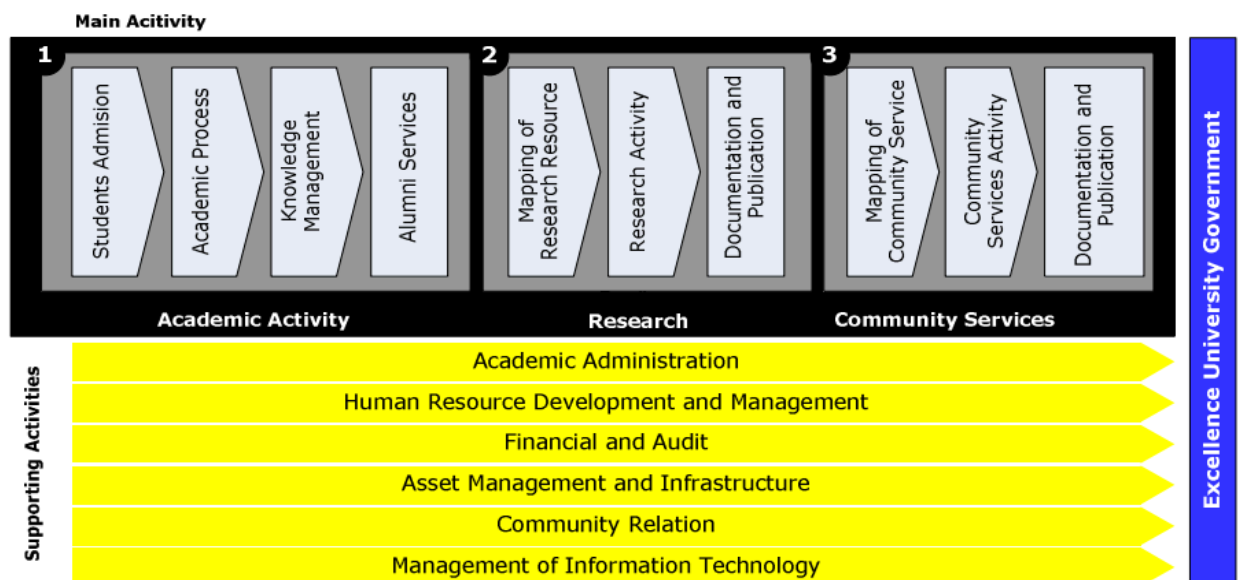


Figure 1.1: Value Chain Analysis at University of Lampung (Nama et al., 2017)

Through an in-depth analysis of the existing architecture, the result indicates that the university's information technology infrastructure was notably fragmented and lacked integration. Key information systems, such as those for academic management, financial operations and human resources were

operating in isolated environments. This separated environment led to inefficiencies, data redundancy, and difficulties in making informed strategic decisions.

In response to these challenges, the study proposes a comprehensive enterprise architecture made up of integrated and modular applications. Significant enhancements include the development of SIAKAD-T, an integrated academic system, alongside the establishment of a centralized library system, unified administrative platforms, a single sign-in solution, a decision support system (DSS) data warehouse, customer relationship management (CRM) tools and improved knowledge management systems (Nama et al., 2017). These solutions are designed to foster integration, support scalability and streamline the university operations.

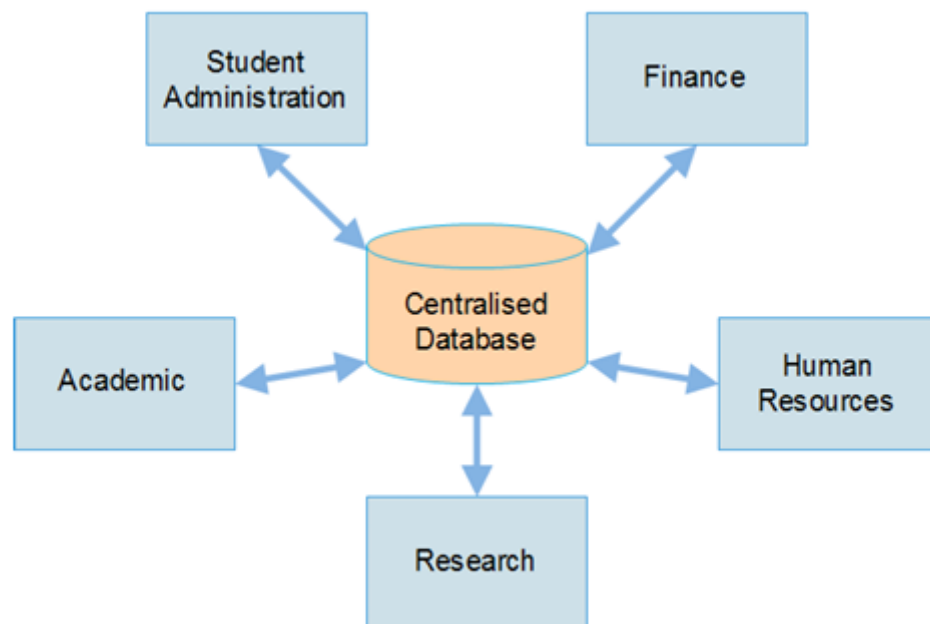


Figure 1.2: Functional Areas of a University Enterprise System at Papua New Guinea Using a Centralised Database (Daniel, 2016)

This paper examined how universities in Papua New Guinea traditionally only operate with isolated information systems that focus on one area such as student admissions, finance and human resources. This more often than not ends up resulting in duplication of data, inconsistent records and inefficient cross-departmental processes. The implementation of enterprise systems was proposed to address these problems, which include using a centralized database and integrated software modules for student admission, enrollment, payroll and procurement. This improved system can enable real-time information sharing and reduce redundancy across departments and modules (Daniel, 2016).

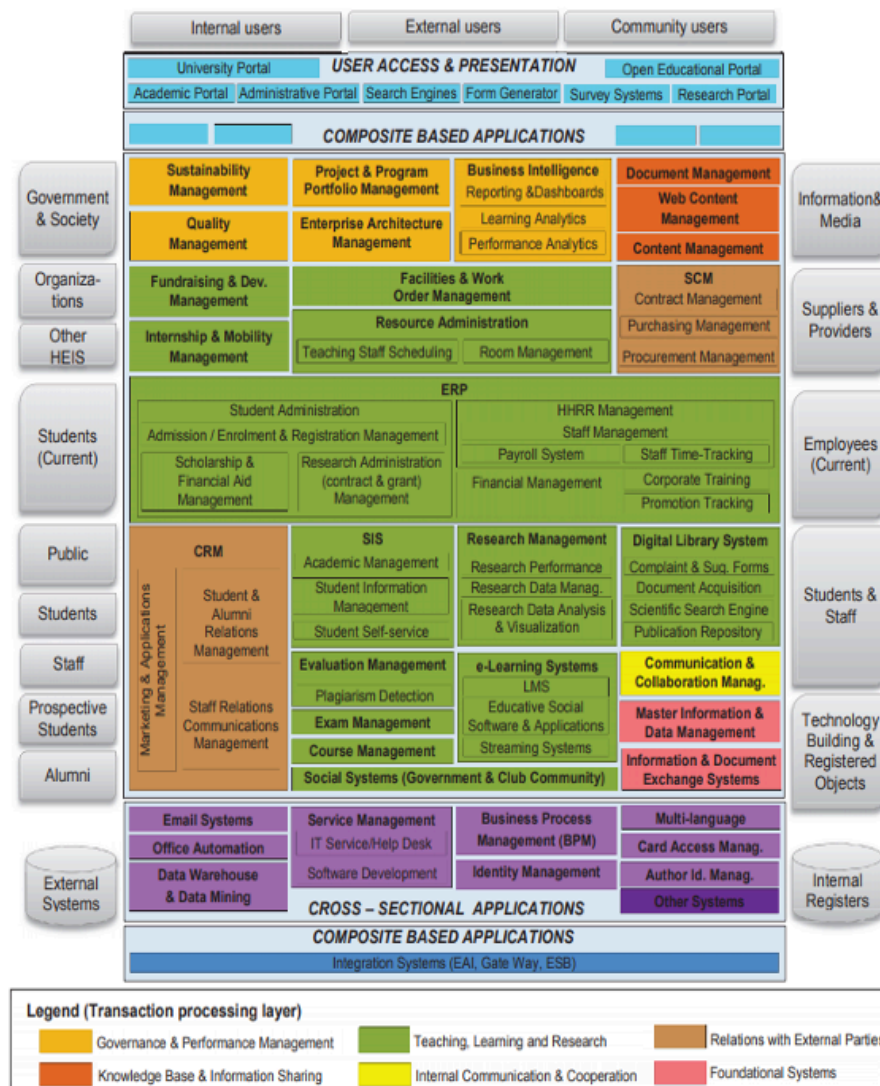


Figure 1.3: Enterprise Architecture Framework for Higher Education Institutions
(Sánchez-Puchol, 2021)

The study provides an in-depth analysis of enterprise architecture in higher education institutes (HEIs) and identifies the challenges faced by these institutions. It is found that most universities struggle with fragmented technological environments, as these often result in inefficiencies when managing student information, academic processes, financial data, and research activities. Fortunately, most of the contemporary universities have leaped into the right tools of combating these problems. More universities are deploying student-administration systems, learning management systems, research management systems and analytics dashboards to ensure everything is in unison. The examples of cases in the paper indicate that as soon as these subsystems are connected together properly, the effects are data becomes more accurate, the load on the administration side reduces, and decision-making becomes more efficient throughout (Sánchez-Puchol, 2021).

2.3 Related Sub-System

This section explains the chosen sub-systems.

2.3.1 Overview of the chosen sub-systems

The chosen subsystems for the University Enterprise Architecture are as follows:

1. Student Information System

This subsystem manages student information including their names, identification numbers, contact details, and emergency contact. It also keeps the academic documentations including admission history, course enrolments, examination results, and academic performances. The system also stores documents such as admission letters, certificates, transcripts and keeps track of graduation requirements.

2. Inventory Management System

This subsystem deals with all the physical assets and resources in the entire university. This encompasses the equipment required in the learning such as computers, laboratory equipment, furniture and teaching aids used in all the faculties. The system manages purchases requests, vendor data and history of purchases. It also handles allocation of assets to all the faculties.

3. Finance Management System

A subsystem that takes care of all financial activities of the organization like the management of student fees depending on the students programs, processing of the tuition fees as well as automatic production of receipts and invoices. It also does payroll on behalf of the staff.

4. Hostel Management System

This subsystem maintains proper documents of the students' hostel residence information and hostel staff information including personal details, room assignments, maintenance and visitor management. The system helps staff control and monitor the daily operations of the hostel.

These 4 subsystems synergistically form a comprehensive university enterprise architecture that supports all major functions needed from a higher education institution. Each subsystem has its own specific functionality but must integrate seamlessly with other subsystems as well to ensure the university's operations can be run efficiently, have consistent data throughout the system and provide sufficient support and meet the university's requirements.

2.4 Technology Used

This section covered main technologies that are used to develop the subsystem that was subsequently incorporated into the University Enterprise Architecture (EA). The development of the subsystem will mainly be centered on the use of SAP Business Technology Platform (SAP BTP). It is an integrated Platform-as-a-Service (PaaS) which provides a unified environment for application development, data management and process automation. SAP BTP is chosen in most of University EA as its ability to lower the gap between legacy enterprise systems and the modern cloud requirements.

2.4.1 Unified Data Management and Analytics

The SAP BTP utilizes SAP HANA Cloud as its primary database technology. This in-memory database allows the university to process numerous amounts of student and administrative data in real-time. It also helps to maintain the structural integrity of complex academic records through advanced multi-model capabilities. Through the SAP BTP service such as SAP Datasphere, the subsystem can easily access all the data from different sources without physically moving the data. This is to ensure “Single source of Truth” across the university’s EA.

2.4.2 Application Development and Integration

The development frameworks within SAP BTP facilitates both Pro-Code and Low-Code environments meanwhile the traditional frameworks like Node.js and Java are supported through the Cloud Foundry environment. This platform emphasizes the SAP Cloud Application Programming Model (CAP) which promotes best practices to ensure the subsystem remains scalable and maintainable. The SAP BTP’s Integration Suite also provides pre-built connectors for enterprise systems to reduce the complexity of API management and allow seamless data flow between departmental applications.

2.4.3 Extension of Enterprise Systems

Core systems such as SAP S/4HANA or Microsoft Dynamics functions to handle sensitive financial and student data in a university environment. The Side-by-Side Extensibility allows developers to build new features for the university subsystem on SAP BTP without the need to modify the code of the core ERP. This is to ensure that the core system inside ERP remains clean, facilitate the future upgrading process and reduce the risk for the institution.

2.4.4 Alignment with EA Frameworks

The SAP BTP platform is designed to support structured EA frameworks like TOGAF and Zachman Framework by providing a standardized platform for four architecture layers which are business architecture, data architecture, application architecture, and technology architecture. This is to ensure the digital processes are aligned with institutional policies and have a consistent governed data handling.

2.5 Summary

On the whole, the literature reviewed in this chapter supports the need for a well-structured and integrated enterprise architecture to ensure the operational challenges of modern universities are addressed. Indeed, findings from previous research have shown that institutions with fragmented, isolated systems are characterized by inefficiencies, inconsistent data, and limited decision-making capability. Enterprise Architecture provides a strategic roadmap for addressing such challenges through clear definition of business, data, application, and technology layers.

Analysis of related subsystems reveals that the core functions of academics and administration, like student records, financial operations, HR, research, and learning management, have to work in harmony for seamless performance at an institutional level. Relational databases, new development frameworks, and enterprise solutions reviewed here provide the required features of scalability, security, and interoperability for the development of a sustainable system. In summary, the literature supports the direction of the design in this project, as it illustrates that the application of EA principles with appropriate technologies will allow the university to advance on aspects of efficiency, integration, and future digitization.

Chapter 3 Methodology

3.1 Introduction

This chapter outlines the methodology that was chosen for the development of UTHM University Enterprise Architecture. The methodology serves as a guide for the approach taken throughout the project's lifecycle. This chapter explains the chosen methodology and the justification for choosing the methodology, an explanation of each phase in the methodology and a project planning schedule is shown to determine the timeline and distribution of tasks of the project.

3.2 Chosen Methodology

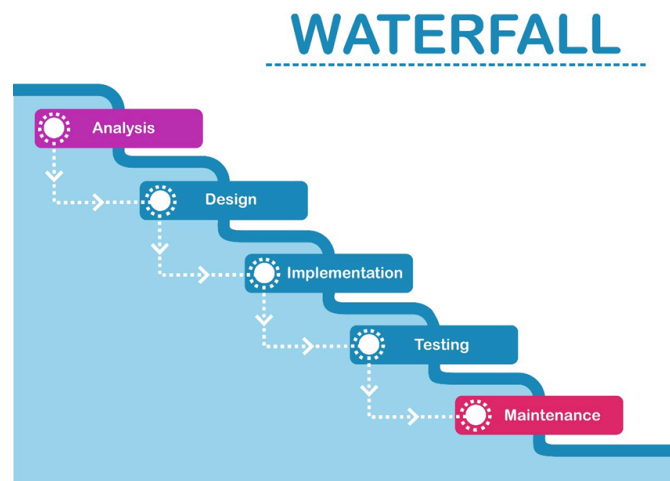


Figure 3.1 Phases in Waterfall Methodology (TechnologyAdvice, 2024)

The chosen methodology for the development of UTHM University Enterprise Architecture is the Waterfall Methodology. Waterfall methodology is a sequential life cycle model that consists of distinct phases which are analysis, design, implementation, testing and maintenance (Senarath, Udesch S., 2021). Each phase has to be completed before beginning the next phase. This methodology was chosen as its clear structure makes it simple and easy to apply to the project development of UTHM University Enterprise Architecture. Using Waterfall methodology also helps with creating a more comprehensive and complete documentation as each stage of the process produces thorough documentation (Samaratova A.K., 2024). Since the Waterfall methodology has clear and well-defined stages, it is easier to estimate timelines and provide more accurate predictions of the duration needed to complete the project.

3.3 Phases of Methodology

The Waterfall methodology for the UTHM University Enterprise Architecture project follows a linear and systematic approach. Each of the phases serves the foundational purpose to ensure the architectural requirements are fully met before moving for the next step.

3.3.1 Planning Phase

Planning phase is the foundation step where it contains all the project's scope, objectives and all the requirements. During the planning phase, the team members gather and do discussion to identify the key stakeholders at UTHM and form a detailed timeline to ensure having a better understanding and strategic planning through the progress. All of the tasks and assignees are gathered using the Gantt chart and GitHub backlog to organize the project roadmap and can track all of the project's progress.

3.3.2 Analysis Phase

The Analysis phase is more focused on gathering and documenting all the required information to complete the project requirements which is the UTHM University Enterprise Architecture (UEA). This phase includes the data collection through the questionnaires that were distributed during the industrial visit at UTHM. All the data collected provides a clear understanding of the current state of their Enterprise Architecture well known as 'As-Is' EA.

3.3.3 Design Phase

During the Design phase, the development of Enterprise Architecture models is going to start. By referring to the requirements collected during the analysis phase, the team designs the 'To-Be' EA together. The specific layer architecture inside the Enterprise Architecture being discussed together. Also, the specific SAP BTP services for example SAP Builds are being used for each subsystem to ensure the subsystem works effectively.

3.3.4 Implementation Phase

Implementation Phase is the phase that creates the subsystem by using the SAP BTP. In this phase, each team member is responsible for developing their assigned subsystem which is the Hostel Management system. This includes the use of SAP BTP cockpit to build the systems and set up the databases on device and create the logic flows specific to its functionality.

3.4 Project Planning Schedule

Project planning schedule is a structured timeline that outlines what tasks need to be done, and when they should be completed so that the project can stay organized and be delivered on time. By following the waterfall methodology as mentioned above, Figure 3.4 shows the gantt chart that needs to be completed for the project.

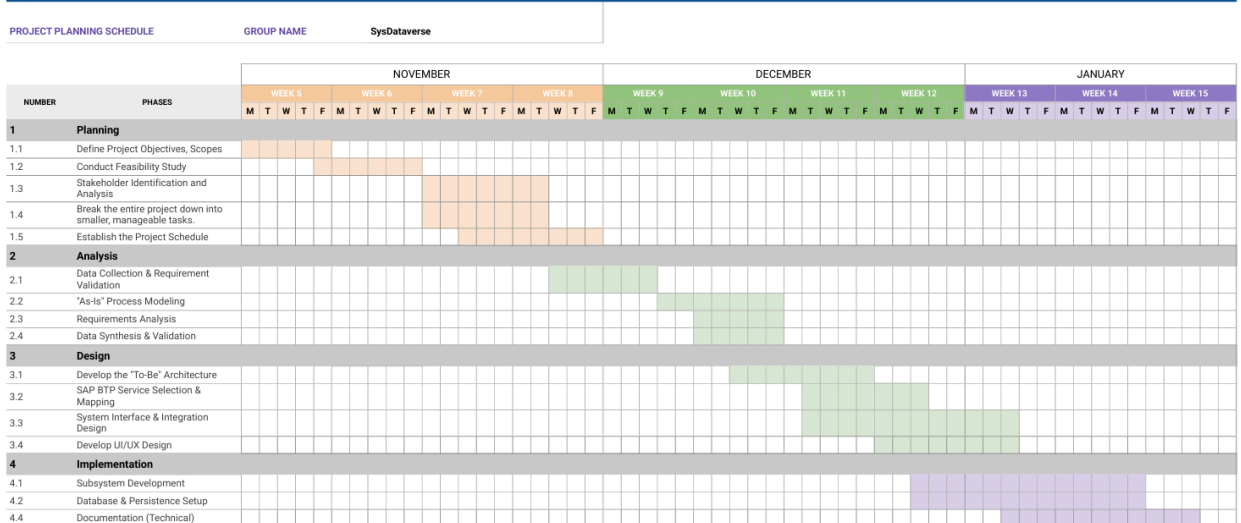


Figure 3.4 Gantt chart of the project

3.5 Summary

The Waterfall Methodology applied in the project are linear and sequential life cycle models that have clear and structured phases. This method was chosen because of its simplicity, clear documentations and ability to give an accurate timeline estimation that are required to complete the task of designing the UTHM Enterprise Architecture. The process of development is separated into 4 phases which are planning, analysis, design, implementation. This is carried out in every stage by breaking down the sub task into a specific time until it is accepted by the stakeholder.

Chapter 4: Analysis and Design

4.1 Introduction

4.2 Organization Structure

4.3 Current System Analysis

The current system operates with all present technological and operational resources which Universiti Tun Hussein Onn Malaysia (UTHM) currently possesses to build its complete digital system. The UTHM digital transformation strategy focused on its goal to create cultural transformation which would lead to digital literacy, better user experience, and modern infrastructure through advanced technology.

4.3.1 Computerized Process and Infrastructure

UTHM digital operations are powered by a modern, web-based architecture leveraging high-availability on-premise infrastructure. The university maintains on-premise data centers designed to Tier 3 standards, featuring dual power sources, Uninterruptible Power Supply (UPS), and generator backups. These critical facilities are located on the upper floors of the data center to protect against environmental risks such as flooding. The university also transformed from a legacy client-server architecture to a new unified system called CENTRAL, which utilizes the web-based technology. This allows lecturers, students and staff to access core modules via a standard web browser by eliminating the accessibility barriers that have been found in the previous system. A centralized master database manages the core student records while the UTHM Learning Management System (LMS), like AUTHOR platform, is responsible for storing all academic materials in a separate database container. These systems are linked through the university-developed APIs and Single Sign-On (SSO) to synchronise students' credentials and monitor in real-time. UTHM also secured their network by utilizing a dual-firewall strategy and Web Application Firewalls (WAF) to defend multi-directional attacks especially from China and Europe.

4.3.2 Manual Processes and Governance

Despite the successful move to web-technology, several workflows still remain manual and semi-automated. Certain general administrative workflows still require the physical completion of forms such as Pusat Kesihatan Universiti (PKU) in UTHM still relies on the traditional manual methods for the patient appointment system. The introduction of any new system or application is strictly regulated by a formal review process as the stakeholders are required to fill a complete Feasibility Study Requirement (FSR) document and need to be approved by the responsible committee. This is to ensure the proposed system is necessary and compatible with the university's goals before the development process.

4.3.3 Identified System Issues

During the student registration for co-curricular subjects, the system expected to handle more than 1000 concurrent users at a time. This peak load performance causes the system to be slowed-down in approximately five minutes of lag times. UTHM also states that the "We Care" support system frequently

receives complaints regarding the forgotten passwords for WiFi authentication and SSO which indicate a need for more streamlined identity recovery.

4.4 Comparison between Existing and Proposed System

4.4.1 Current Digital Infrastructure

UTHM has established a comprehensive digital foundation built upon industry-standard technological components. The university operates a Tier-3 standard data center with modern web-based architecture and multi-layered security protocols, including dual firewalls and Web Application Firewalls. The centralized information system (CENTRAL) and learning management system function as integrated platforms through university-developed APIs and Single Sign-On capabilities.

This infrastructure has enhanced digital accessibility across the institution. However, several operational workflows remain dependent on manual processes, particularly administrative documentation and academic advisor appointment scheduling systems.

4.4.2 Operational Challenges

The current system exhibits notable performance limitations during high-demand periods. When concurrent user access exceeds 1,000 individuals, such as during co-curricular subject registration, the system experiences significant performance degradation, resulting in approximately five-minute response delays.

Additionally, the "We Care" support channel receives persistent password-related inquiries, indicating inadequate user identity recovery mechanisms within the existing framework.

4.4.3 Proposed System Enhancement

The proposed digital ecosystem addresses these operational deficiencies through comprehensive infrastructure improvements. The enhancement plan includes advanced load balancing mechanisms, performance optimization protocols, and potential hybrid cloud integration to accommodate increased concurrent usage.

Key development areas include:

- 1) **Complete workflow automation:** Digital transformation of remaining manual processes, including online academic advisor appointments and electronic administrative forms
- 2) **Streamlined governance processes:** Implementation of fully digital approval workflows, such as online FSR submissions and automated committee procedures
- 3) **Enhanced user experience:** Development of a unified platform interface with improved identity management capabilities

- 4) **Advanced security features:** Integration of self-service password recovery systems and multi-factor authentication options

4.4.4 Expected Outcomes

The proposed system represents a significant advancement from the current semi-digitized infrastructure. The enhanced platform will deliver improved scalability, comprehensive automation, strengthened security protocols, and optimized performance capabilities. This transformation will create a more cohesive digital environment that better serves the university community's operational and academic needs.

The transition from the existing system to the proposed infrastructure reflects a strategic move toward a fully integrated, performance-driven digital ecosystem that can effectively support the university's growing technological demands.

4.5 System Requirements Gathering Techniques

We conducted an interview session in the industrial talk to gain a deeper understanding of UTHM's daily operations, enterprise architecture, and the practical challenges faced by the ICT team and users.

The questions are categorized below:

4.5.1 Daily Operations & Current Process

1. Can you describe the main tasks you perform daily using the current system?
2. Although APIs have been adopted to integrate the data, there is still manual transfer of data between the Main Student Database and the LMS. What are the technical or procedural obstacles that would not allow total automation?

4.5.2 Enterprise Architecture & Integration

1. Now that the migration of the legacy TCIS to the web-based CENTRAL architecture has reached completion of 70-80% is already made, what do you believe are the largest technical challenges that could be expected to be encountered during the remaining 20%?
2. Given that the LMS is designed to be independent and exist as a database separate on the central system, how can data consistency and integrity be ensured so that one cannot find out inconsistency including the absence of student records?

4.5.3 Infrastructure & Performance

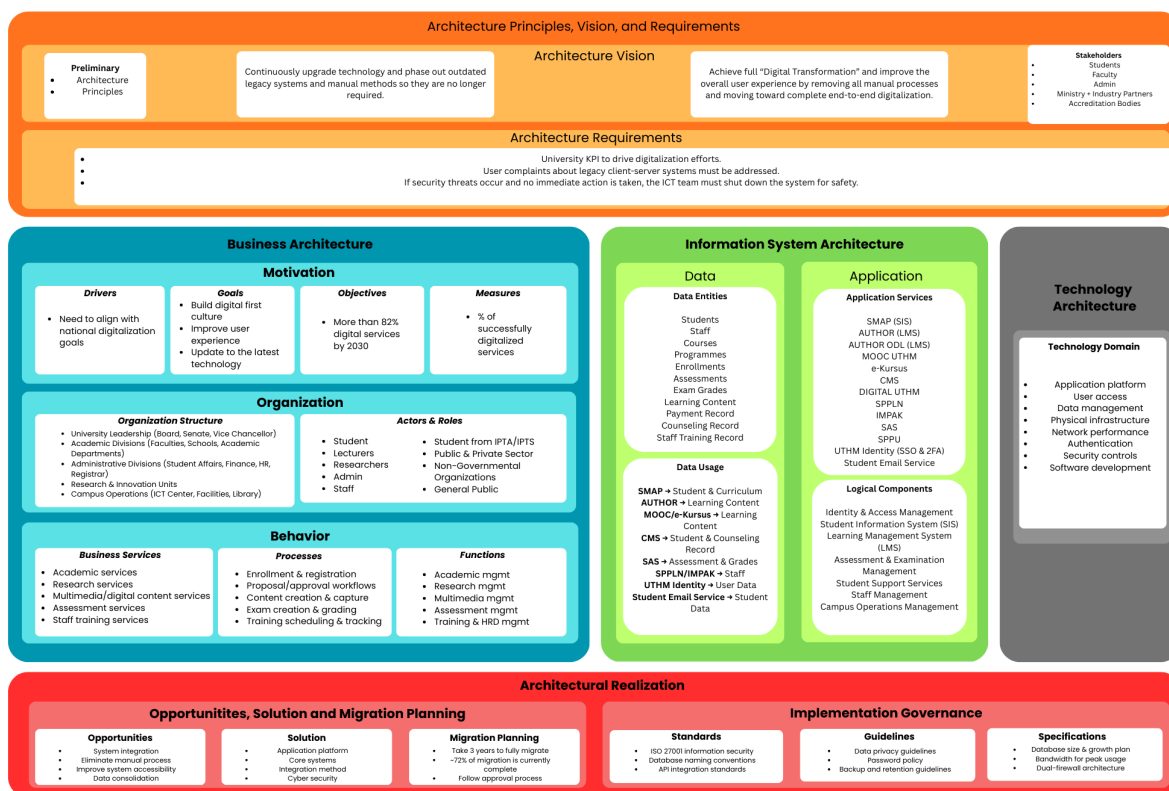
1. The latency was observed to have a time of 5 minutes during peak Course Registration (CRS) intervals that have more than 1000 simultaneous users. What is the solution to this problem by the UTHM team?
2. You explained the 3 tier power backup (TNB sources A/B, UPS, Genset). How much time will the existing UPS solution support the critical servers until the genset goes online to replace all its power?
3. Why is the Data Centre located upstairs despite the logistical convenience of a ground-floor location for loading and unloading equipment?

4.5.4 User Experience & Future Needs

1. How does the cybersecurity team protect sensitive data from cyberattack?

4.8. System Design

4.8.1 Enterprise architecture EA-AS IS



4.8.2 Explain each component in system architecture EA-AS IS

1. Architecture Principles, Vision and Requirements

a. Preliminary Principle

The main order of action is focused on gradually upgrading technology and phasing out outdated legacy systems and manual methods. This is the current transition stage of UTHM in which the institution is transitioning out of an old client-server architecture to modern web-based architecture, which is still ongoing.

b. Architecture Vision

The current vision emphasizes achieving full digital transformation by improving the overall user experience through the removal of manual processes and moving toward complete digitalization. At present, UTHM is at approximately 70-80% digital migration, meaning there is still a significant portion of operations that rely on manual or semi-automated processes.

c. Architecture Requirements

The AS-IS requirements focus on three critical areas:

1. University KPI alignment: Ensuring systems support the institution's key performance indicators and digitalization goals
2. Legacy system user complaints: Addressing issues users face with outdated client-server systems that require specific configurations
3. Security protocols: If security threats occur and no immediate action is taken, the ICT team must have protocols to shut down the system for safety

2. Business Architecture

This layer defines the existing posture of UTHM in its operations in terms of business, including organisational structure, processes, and stakeholder interactions.

A. Motivation Layer

Drivers: The existing system is driven by the need to comply with the national digitalisation goals that are set by the Ministry of Higher Education. UTHM is aware of the fact that the outdated technology and manual procedures limit the level of operational performance and undermine user experience.

Goals:

1. Create a digital-first culture within the institution.
2. Improve the experience of both faculty and students.
3. Improve infrastructure using the latest technology to ensure that the company is up to date.

Objectives: Its main agenda is to achieve over 82-percent digital services by 2030 and thus shifting paper-based and manual workflow to complete automated digital services.

Measures: Success is currently measured using the level of services that have been successful in digitalisation. UTHM keeps track of the volume of processes moved on-line and the adoption of the new web-based CENTRAL system by the users.

B. Organization Layer

The current organisational structure of UTHM is made of:

- University Management: Vice Chancellor, Board of Directors.
- Functional Divisions: Organized in terms of faculties, schools, and academic departments.

- Administrative Departments: Student Affairs, Finance, Human Resources and Registrar departments.
- Support Units: Research and Innovation Unit, ICT Centre, Facilities management, and Library.

Actors & Roles:

There are several stakeholders that the system is serving at present:

- Students: For IPTA (public universities)/IPTS (private institutions): primary users.
- Academic members of staff who present instructional content are known as lecturers.
- Researchers: Faculty taking part in scholarly research work.
- Administrative Personnel: Personnel who are involved with operational management.
- Personnel: General support staff.
- External users: partners in the public and non-governmental organisations as well as the general population to receive stated services.

C. Behaviour

The business services already available are:

- Academic services (enrolment, registration, grading).
- Research services (project management, grant tracking).
- Student services (counselling, support services).
- Evaluation services (examinations, evaluations).
- Staff trainer services (professional development).

Processes: Some of the major business processes that are currently in operations are enrollment and registration, proposal approval, content development and acquisition, exam construction and scoring, training organization and monitoring.

Functions: The fundamental functional areas include course scheduling, planning curriculum., research management, multimedia control, evaluation administration, staff development through training and HRD management.

3. Information System Architecture

A. Data Architecture

In data architecture, the core data and how these data relate to the others in information systems of UTHM are carefully established. Students, Staff, Courses, Programmes, Enrollments, Assessments, Grades, Learning Content, Payment Records, Counseling Records, and Staff Training Records are some of the most significant academic and administrative entities present in the existing AS-IS environment. These organizations in totality support the academic delivery, student services and the overall operations of the institution.

This architecture makes the utilisation of the data easier through various systems that exchange and receive data to protect the continuity of operation. Student Management and Academic Planning (SMAP) is the key source of information on students and curriculum related information. The learning materials are delivered and shared on AUTHOR, MOOC UTHM and e-Kursus. The Content Management System (CMS) handles the student documentation and the data relating to counseling. The results of the assessment and records of the academic performance are stored in the Student Assessment System (SAS). Information related to staff is monitored by SPPLN and IMPAK and the authentication of users is centralised by UTHM Identity, therefore, facilitating Single Sign-On (SSO). Moreover, there is provision of Student Email Service that is used to facilitate official communication associated with student records.

B. Application Architecture

Application architecture outlines the essential systems that will be implemented to assist UTHM academics and administration processes. This includes primary applications of SMAP as the Student Information System (SIS), AUTHOR and AUTHOR ODL as Learning Management Systems (LMS), and additional applications, including MOOC UTHM and e-kursus to deliver long-distance learning. CMS, DIGITAL UTHM, SPPLN, IMPAK, SAS, and SPPU are all the administrative and support functions that are facilitated. UTHM Identity imposes identity management and secured access complemented with institutional email services.

Logically, the applications are grouped in functional components, which are Identity and Access Management, Student Information Management, Learning Management, Assessment and Examination Management, Student Support Services, Staff Management and Campus Operations Management. All these elements together form a unified ecosystem that supports the main academic and working processes at UTHM.

5. Technology Architecture

The technology architecture of UTHM mainly focuses on transitioning from a legacy client-server model to a modern, fully web-based platform. This is to ensure that their technology has ubiquitous access for students and staff in the university. UTHM has a high-availability infrastructure that is built on a very solid foundation. It has three internal data centers that place compliance and internal control above cloud

services. In an order of priority for business continuity, UTHM has put in place a three-tier power redundancy system made up of a dual-power source, Uninterruptible Power Supplies (UPS) for immediate backup, and Gensets for long-term outages. Security, being an essential factor, UTHM has adopted a multi-brand dual-firewall architecture and Web Application Firewall (WAF) which altogether protect the institutional portals. This layer also standardizes API integration and Single Sign-On (SSO) to facilitate a seamless communication between the centralized core database and peripheral learning systems like AUTHOR.

6. Architectural Realization

This layer explains the opportunities, solution, and migration planning, and the implementation governance that UTHM has implemented.

A. Opportunities, solutions, and migration planning

Opportunities: This section identifies the key improvements that UTHM must enable to derive from their strategic goals and identified problems. The institution focuses on the system integration and eliminating the manual processes to reduce the administrative work, aiming for an End-to-End (E2E) digital goal by 2030. UTHM also prioritizes improving their system accessibility by replacing the legacy client-server architectures with fully web-based systems and ensuring their data consolidation by maintaining a single, centralized database linked to separate systems like AUTHOR.

Solutions: The solution defines the final desired state for UTHM's technology, application and data architecture. The application platform must be fully web-based for ubiquitous access, while core systems utilize a centralized system for the core data and AUTHOR as the LMS. The integration method of UTHM relies on the API and SSO supported by the UTHM data center infrastructure which focuses on the internal infrastructure and compliance. For reliability, UTHM maintains a 3-tier power redundancy and a multi-brand security layer featuring dual firewalls alongside a Web Application Firewall.

Migration planning: This outlines the strategy to execute the transition to the target state. UTHM has utilized a "face-to-face" migration strategy where the old system runs concurrently with the new implementation. The UTHM has estimated their migration duration is about 3 years as they have reached approximately 72% of completion. All new applications must strictly follow the UTHM approval process including filling out a complete MSR requirements and committee check.

B. Implementation governance

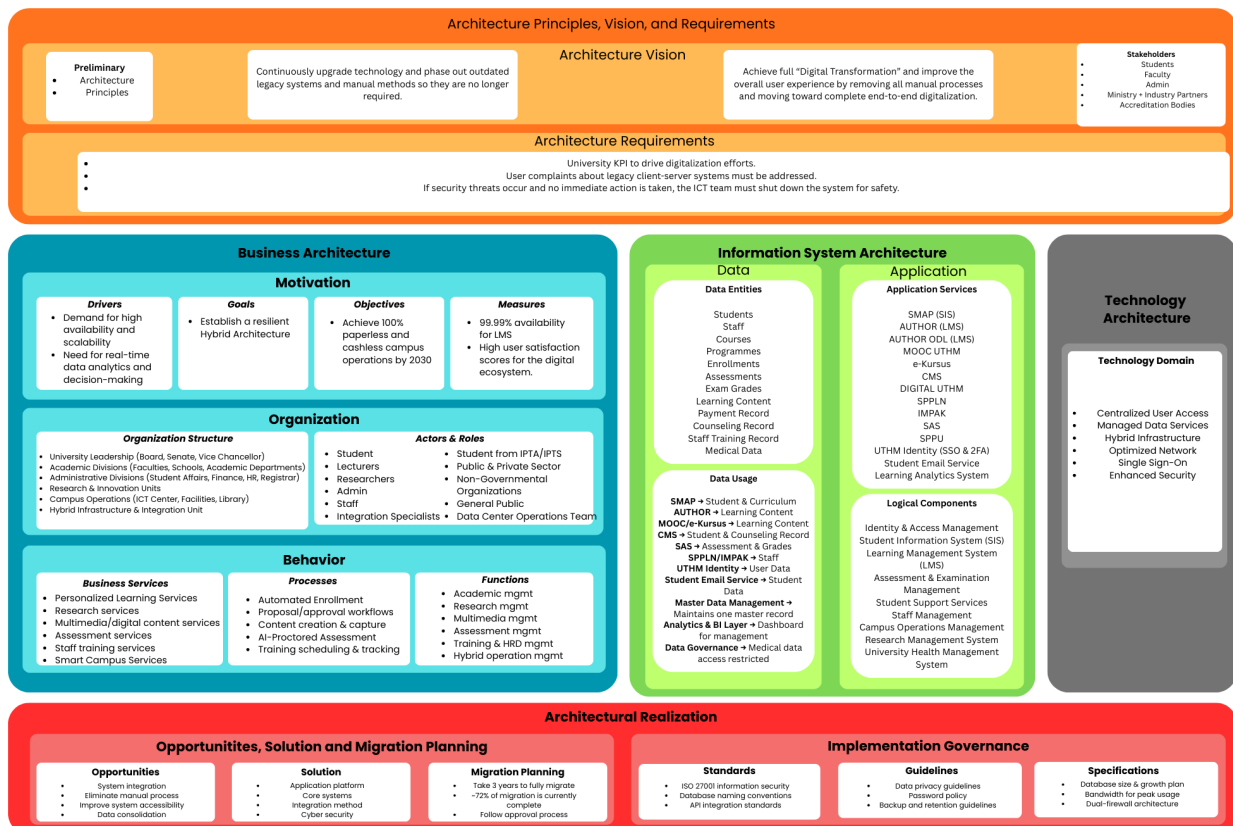
Standards: This component is to define a formal, mandated criteria to ensure the quality and

consistency across UTHM systems. The ISO 27001 information security helps the UTHM to formalize the commitment to robust their security systems. UTHM also mandates the database naming conventions to manage the centralized but segmented data environment and API integration standards to ensure the smooth data transfer between the core database and the LMS.

Guidelines: Data privacy guidelines helps the UTHM to ensure compliance with PDSA standards for sensitive student data and academic marks. A structured password policy is established to address user experience issues meanwhile the backup and retention guidelines are implemented in UTHM to protect the centralized data and ensure infrastructure continuity.

Specifications: These include the database size and growth plan to ensure the centralized system does not degrade as the university moves toward 2030 digital goals. The UTHM has also addressed their bandwidth for peak usage specifically targeting bottlenecks that have been observed during the student co-curricular registration.

4.8.3 Enterprise architecture EA-TO BE



4.8.4 Explain each component in system architecture EA- TO BE

The elements of a TO-BE architecture of a university are usually based on the TOGAF layers but are directed to optimization and future possibilities. There are some of the components in the EA TO-BE:

1. Architecture Principle, Vision, and Requirements

This layer determines the objectives of the transformation at the high level.

i. Architecture Principle:

Architecture Principles are the rules and guidelines, which guide UTHM in developing and managing its IT resources. The principles are focused on interoperability, which will mean that different systems such as Finance and Academic records can communicate with each other without issues, and agility, which will enable the infrastructure to expand as the university grows. These principles are focused on a user-centric concept and they make sure that all the technological developments are meant to enhance the performance and experience of students and staff instead of complicating their day to day activities.

ii. Architecture Vision:

The Architecture Vision serves as the North Star to the university and is the description of the way it wants to be fully integrated into Digital UTHM. The first objective is to become 82% digital by 2030, to move to the modern digital ecosystem, frontrunner of fragmented legacy systems and paper-based workflows. The vision will focus on ensuring that there is a single experience where all stakeholders can use a single secure portal to access all the university services and create a culture of data-drivenness that will ensure the mission of excellence of TVET education is achieved by the university.

iii. Architecture Requirements:

Architecture Requirements are measurable, specific, "must-haves" which the system has to fulfill in order to be declared successful. They comprise a high level of compliance with security measures, including the use of Multi-Factor Authentication (MFA) and ISO 27001 standards to keep safe institutional data. Also, these requirements will guarantee that the IT strategy is optimally aligned with the Ministry of Higher Education (MOHE) KPIs and directly responds to user pain points, as the new applications should be stable, high-performing and address the real needs of the campus community.

2. Business architecture

This layer justifies the motivation, organization, and behaviour required in the future Hybrid Architecture of UTHM.

A. Motivation Layer

Drivers: The university is under pressure by external forces to offer systems that are online at all times and capable of expanding with the needs of the students. The management team must also have real-time data analytics that will enable them to make decisions instantly rather than depending on manual reports which are very slow. These drivers push UTHM out of simple web sites to robust, scalable web sites.

Goals: The first objective is to create a Hybrid Architecture that will integrate the security of on-premise servers. This guarantees safety of UTHM data and still produces the same efficiency and accessibility of academic applications.

Objectives: UTHM is also aiming at going 100 percent paperless, cashless by the year 2030. This will mean the elimination of all the physical forms of operations and cash transactions to smoothen the day-to-day operations. This will save time and efficiency of administration.

Measures: The success measure will be to achieve the 99.99 percent system availability, this implies that the Learning Management System (LMS) is practically never offline. The other key sign is the high user satisfaction rates and it proves that the new digital ecosystem is not a waste of time to students and employees. These steps imply that the technology will offer real value and not new features.

B. Organization Layer

In order to facilitate the new hybrid architecture, we have introduced a new Hybrid Infrastructure and Integration Unit to take care of the key relationship between the on-premise servers at UTHM and the cloud. This department will work alongside current departments but will add new professional skills such as Integration Specialists whose duty will be to make sure that the system is not only interoperable but also with a Data Center Operations Team that focuses on the physical Tier 3 infrastructure.

C. Behaviour

Business Services: We incorporate Personalized Learning Services whereby the content is adjusted to the needs of individual students rather than having a general lesson. Besides that, we include a new service known as Smart Campus Services which uses IoT technology to access such facilities as smart parking and smart libraries. This changes the emphasis on the delivery of basic services to intelligent and user-friendly experience.

Processes: We replaced the manual processes with Automated Enrollment whereby we could regulate instantly without the delay of the human approval. The introduction of AI-Proctored Assessment is also aimed to provide the control of online examinations, which are autopiloted and

avoid bodily invigilators. These procedures are more focused on expediency and automation as opposed to manual data input.

Functions: Hybrid Operation Management has now been incorporated into the management functions in order to manage the cloud services and the on-premise hardware at the same time. This provides a single perspective of the overall IT health of the university. It enables IT teams to easily and proactively manage problems in various environments.

3. Information System Architecture

The Information system architecture level connects between the business strategy at the high level and the technology. It consists of two major fields: Data Architecture that describes what needs to be known, and Application Architecture that describes the computer programs that manage that information.

a) Data architecture

Data architecture involves defining important data objects and the flow of information within the organization in order to maintain data integrity and security.

The most basic units of the UTHM information ecosystem are data entities. It could be subdivided into their particular role:

- Essential Academic Branches: Students, personnel, courses, and programs.
- Academic history: classifications, examinations, and grades.
- Operational Assets: Learning material and payment history.
- Sensitive Records: Special training and medical history records.

The movement of data between systems is called data usage. It involves the data being utilised to bring value to the various stakeholders.

- LMS Integration: Learning Management Systems draw on student and curriculum information to load the online classrooms.
- Administrative Management: Content Management Systems (CMS) is used to align student records with learning materials.
- Security & Identity: Identity systems are used to control credentials of users to make sure the correct individuals can access the correct data.
- Strategic Intelligence: Business intelligence (BI) tools combine data to provide a management dashboard.
- Governance: The medical data access is highly divided and limited to the privacy policies.

b) Application architecture

Application architecture is a concept that is used to describe the software services and logical modules, which facilitate user interaction with the data.

Application services is the functional software platforms deployed:

- Learning Platforms: SMAP, AUTHOR and MOOC LMS suites.
- Administrative Platforms: CMS and e-Kursus in managing the curriculum.
- Enterprise Services: Learning Analytics, Identity management and Email systems.

Logical component represents the central driver of the digital operations of the university:

- Identity & Access Management: Authentication and user rights.
- Student Information System (SIS): This is the database of student lifecycle.
- Learning Management (LMS): The teaching, learning, and collaboration interface.
- Assessment and examination: Intensity course modules: grading and exam integrity.
- Campus Operations: Systems: Research and day-to-day logistical systems.
- Health Management: A safe university medical service system.

In general the comparison between the information system architecture of EA as-is and EA to-be is that the To-Be architecture added completely new modules that the As-Is does not have, namely the University Health Management System. Then, implements a special Master Data Management layer and Data Governance framework to manage limited access, especially of medical records. Finally, includes an Analytics and BI Layer to convert the raw data into actionable dashboards to the university management.

4. Technology Architecture

This layer defines the software and hardware infrastructure required to support the deployment of business, data, and application services. The TO-BE technology architecture for UTHM shifts from fragmented, legacy hardware to a resilient, high-performance hybrid environment.

i. Hybrid Infrastructure

The core of the TO-BE model is a Hybrid Cloud environment. This combines UTHM's on-premise Tier 3 Data Center for sensitive localized data (like medical and financial records) with Scalable Cloud Services such as AWS or Azure for student-facing applications like the LMS and MOOC. This ensures the university can handle massive traffic spikes during enrollment or exam periods without system crashes.

ii. Centralized Architecture & Managed Data Services

Unlike the previous decentralized "silo" approach, the TO-BE architecture utilizes Centralized Management. All server resources, storage, and databases are governed under a unified framework. Managed Data Services are implemented to automate backups, database tuning, and high-availability clustering, ensuring that the "Master Data" remains consistent across all campus departments.

iii. Optimized Network & Connectivity

To support the "Smart Campus" and IoT goals (smart parking/libraries), the network infrastructure is upgraded to High-Density Wi-Fi 6 and a 10Gbps Fiber Backbone. This optimization ensures low latency for AI-proctored assessments and seamless connectivity for thousands of simultaneous mobile users across the UTHM campus.

iv. Single Sign-On (SSO) & Identity Management

To solve login fatigue issue faced by students and staffs, the technology layer implements a Unified Identity Provider (IdP). Using protocols like SAML or OpenID Connect, the Single Sign-On (SSO) allows students and staff to access the LMS, Finance, HR, and Medical systems using a single set of UTHM credentials, significantly improving the user experience and reducing password-reset tickets.

v. Enhanced Security (Zero Trust & Dual-Firewall)

Security is integrated directly into the infrastructure through a Zero Trust Architecture.

- Dual-Firewall Design: Traffic is inspected through both external perimeter firewalls and internal micro-segmentation firewalls to protect sensitive data zones.
- Encryption at Rest & in Transit: All data moving between the cloud and on-premise servers is encrypted using TLS 1.3, and databases are encrypted to meet ISO 27001 standards.

5. Architectural Realization

i. Opportunities & Solutions:

Opportunities and Solutions are the next phase where UTHM establishes the What by assessing the disconnect between the existing legacy systems and the envisioned new digital future. This element is aimed at bundling together similar business requirements into rational work packages i.e. consolidating the manual leave and claims processes into one "HR Digital Module." Within the framework of this project, it explains the manner in which UTHM decides on the outdated

client-server applications that need to be removed or upgraded to achieve its 82% digitalization goal in the university.

ii. Migration Planning:

Migration Planning determines the "When" through creating a time-based roadmap of IT transformation of the university. As shown in the EA diagram, UTHM is at stage of polishing with 72 percent of the migration already accomplished as part of the 3-year strategic plan. The component is aimed at having digital projects prioritized effectively depending on their available budget as well as the urgency of security threats and accomplishment of critical university KPIs, so that there is a smooth transition to the target architecture.

iii. Implementation Governance:

The Implementation Governance is the How by implementing the rules and standards that make sure that new systems are safe and well-integrated. UTHM is employing ISO 27001 as a reference to the information security benchmark and fully demands technical control by means of specifications, such as the so-called Dual-Firewall design. This element plays a policeman role, making sure that new development is done to the standards of the institution, such as requiring the use of UTHM Identity (SSO/2FA), so as to not create an insecure or fragmented system of logging in.

4.8.3 System Architecture

The Cloud System Architecture diagram relies on the SAP Business Technology Platform (BTP). The design uses a Low-Code/No-Code strategy to ensure efficiency and scalability. The system comprises four key layers. The Application Layer leverages SAP Build to handle core logic like room allocation. The Data Layer centralizes all information using device storage to store records. Finally, the Analytics Layer uses SAP Analytics Cloud to visualize key metrics through dashboards. This setup ensures a complete workflow from user input to strategic reporting.

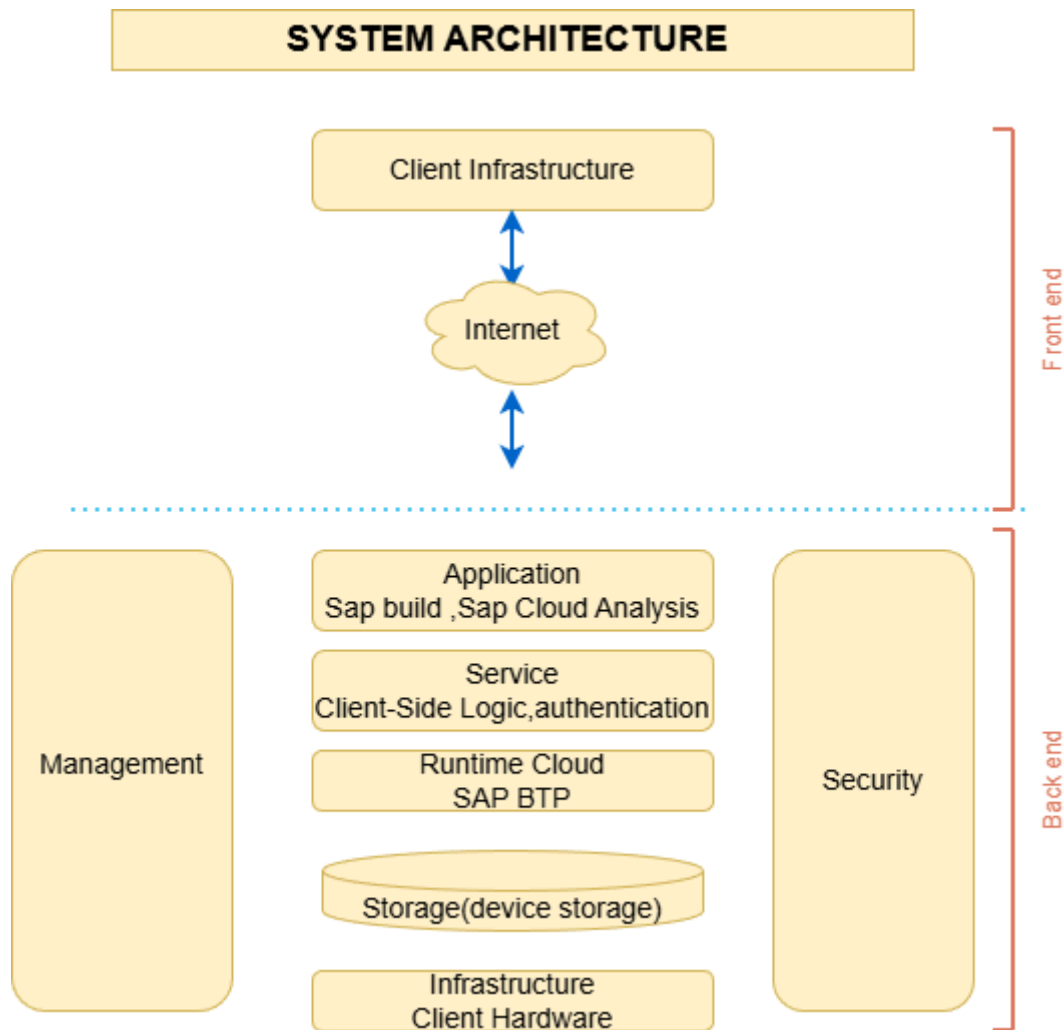


Figure 4.8.3.1 system architecture of hostel system

4.8.4 System Architecture Component Explanation

It is segregated into 2 broad parts; Front End (Client-side) and Back End (Server-side).

Front End (Top Section)

This is the part that the user interacts with the system.

Client Infrastructure This describes the environments and devices the end-users are operating (e.g., laptops, smartphones, web browsers). It acts as a point of entry to the user in the application.

Internet: The medium of communication. It links the Client Infrastructure and the Back End services and data flows in both directions (which are indicated by the blue double-headed arrows).

Back End (Bottom Section)

This is the nub of the system on which processing, logic and data storage occur. It is surrounded by two vertical pillars (Management and Security) and a technology stack at the center.

These layers are organized in a downward manner starting with the software application to the physical hardware.

Application:

SAP Build: It indicates that the application was built on the low-code/no-code environment of SAP. It enables the development of enterprise apps in a short time.

SAP Analytics Cloud: this element manages data visualization, reporting and business intelligence (BI) in the app.

Service:

Client- Side Logic: Although it is written on the Client- Side the fact that this is placed in the backend indicates that this layer handles the serving or coordination of business rules which determine how the client will behave.

Authentication: This service is a service that identifies user identities before they are allowed to access the application hence only authorized users can access the application.

Runtime Cloud:

SAP BTP (Business Technology Platform): It is the core of the architecture. SAP BTP is a Platform-as-a-Service (PaaS) which is used to host the applications, carry out integration, and offer environment where the apps can be executed.

Storage (Device Storage):

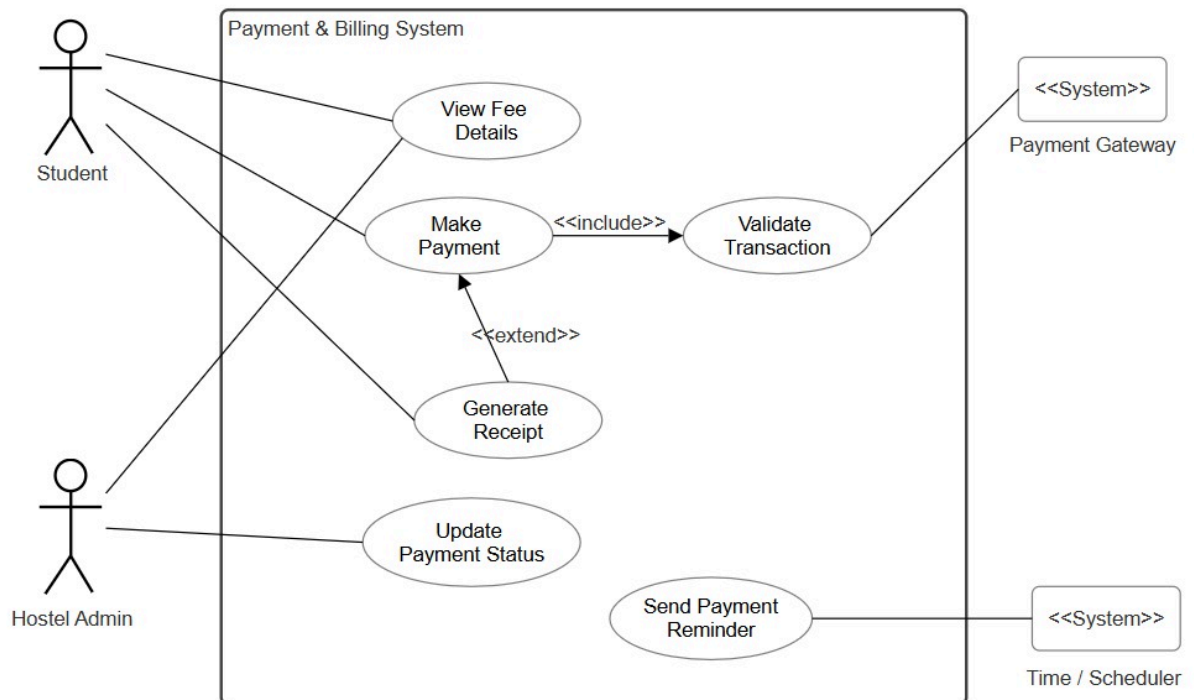
This is the persistence layer and data is stored in the form of a database cylinder. The term device storage normally refers to the actual physical or virtual disk space in which the data are stored.

Infrastructure:

Client Hardware: This is the physical computing hardware that is used in supporting storage and runtime environments.

4.8.5 Project Design

4.8.5.1 Use Case Diagram - Payment & Billing Tracker



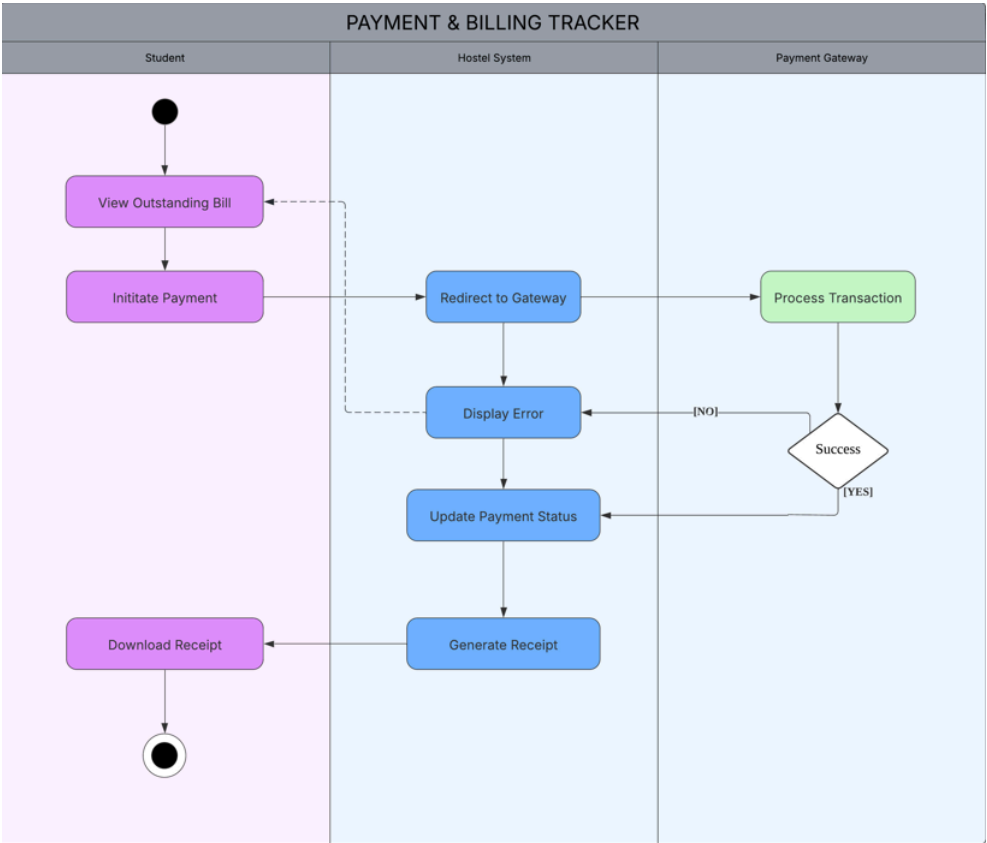
4.8.5.2 Use Case Description - Payment & Billing Tracker

Use Case Name		Payment & Billing Tracker
Actor		Student, Hostel Admin
Preconditions		1. The student must be logged into the Hostel Pay system. 2. An outstanding bill must exist for the student.
Normal Flow	Description	1. Students navigate to the "Hostel Bills" section. 2. The system displays the current outstanding balance and bill details. 3. Students initiate the payment process for the specific bill. 4. System validates the request and redirects the connection

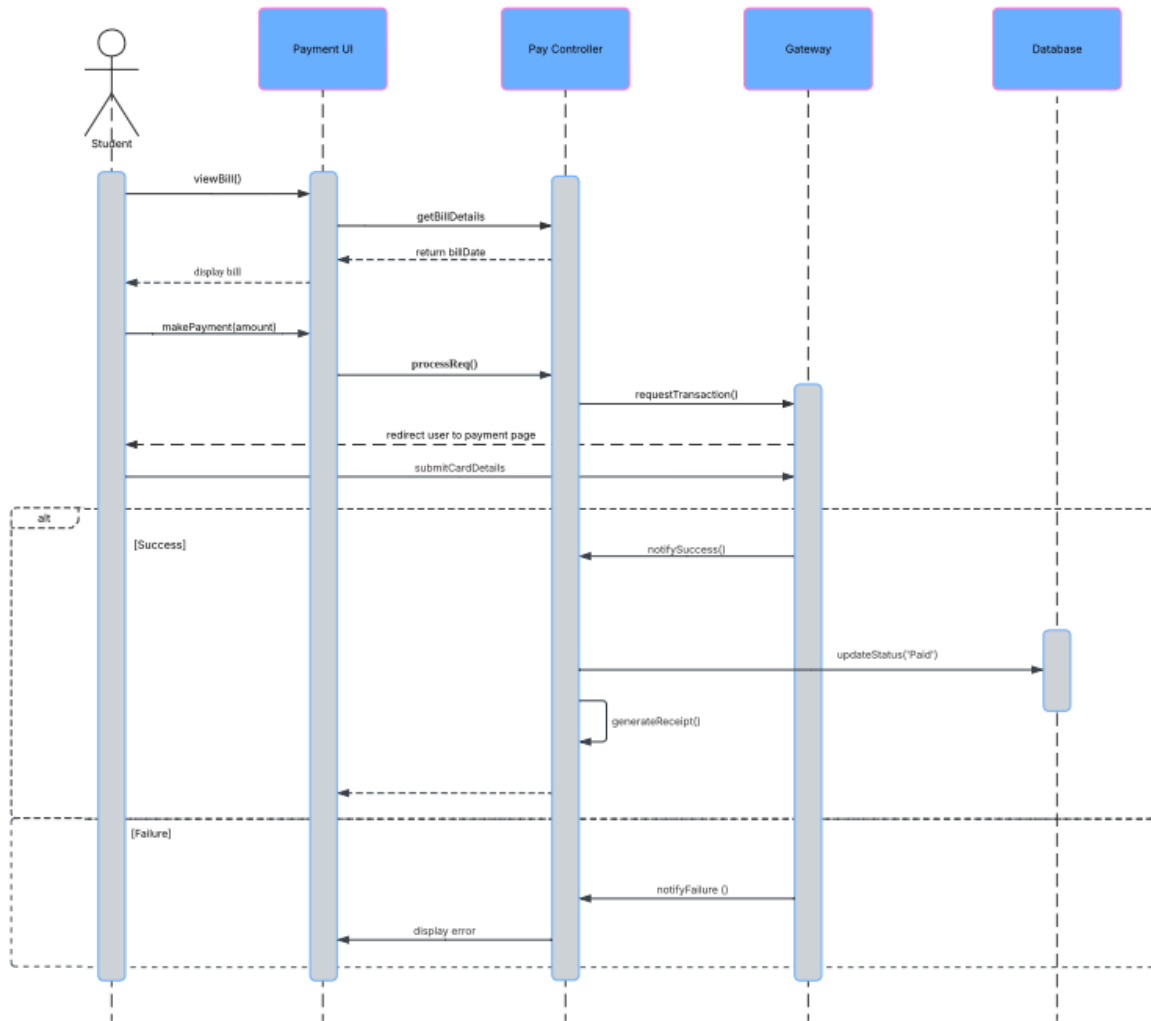
		to the external Payment Gateway. 5. Payment Gateway processes the transaction securely. 6. Payment Gateway confirms the successful transaction to the System. 7. The system records the payment and updates the bill status to "Paid". 8. The system generates a receipt for the transaction. 9. The system notifies the Student that the payment was successful.
	Postconditions	Success: The Student's outstanding balance is updated to 0, the Billing status is set to 'Paid', and a new Receipt record is created. Failure: The database remains unchanged, and the bill remains pending.
Alternative flows		A1. Payment Failed / Rejected - At Step 4, if the Payment Gateway encounters an error (insufficient funds, wrong OTP, cancelled). - The Gateway sends a notifyFailure() signal to the System. - The System displays a "Transaction Failed" error message on the UI. - The System redirects the Student back to the Dashboard. - The Bill status remains "Overdue" or "Unpaid". A2. Gateway Timeout - If the Student stays on the Gateway page for too long without action. - The session expires, and the Student is returned to the app with a "Session Timeout" notification.
Non-functional requirements		1. Security: Data encryption using TLS 1.3 for secure transmission and compliance with PDPA standards to protect student financial information. 2. Performance: The system should load the payment page within 2 seconds and support up to 1000 concurrent users

	during peak periods. 3. Usability: The interface must be fully responsive for mobile devices and provide immediate, clear feedback messages to the user.
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4.8.5.3 Activity Diagram - Payment & Billing Tracker

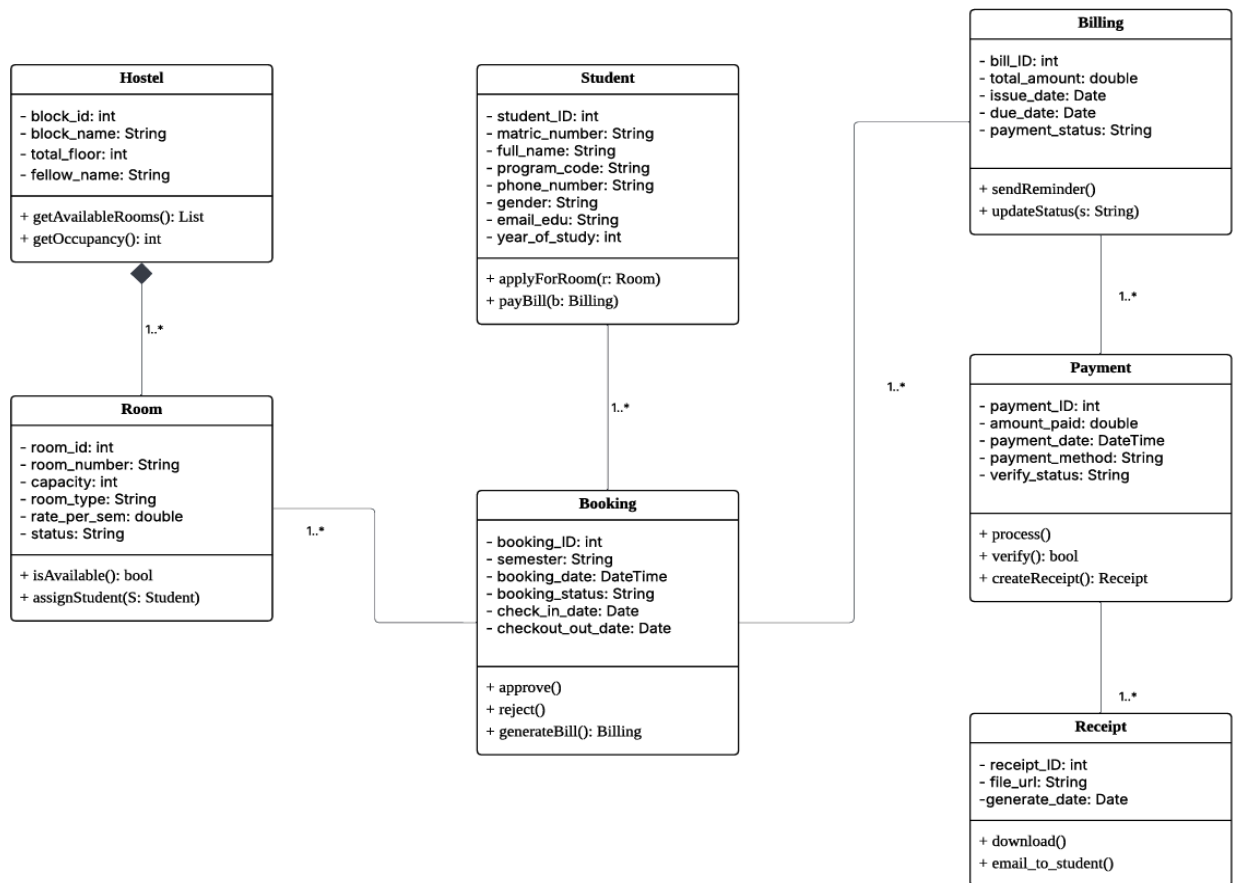


4.8.5.4 Sequence Diagram - Payment & Billing Tracker

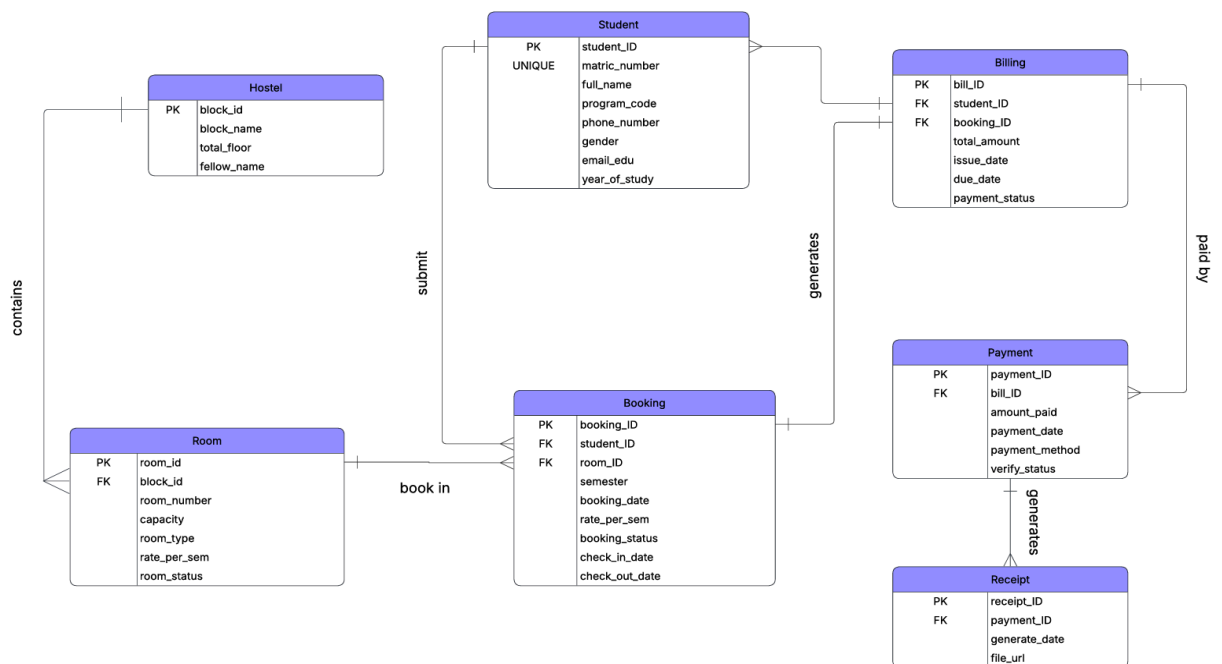


4.8.6 Database Design - Payment & Billing Tracker

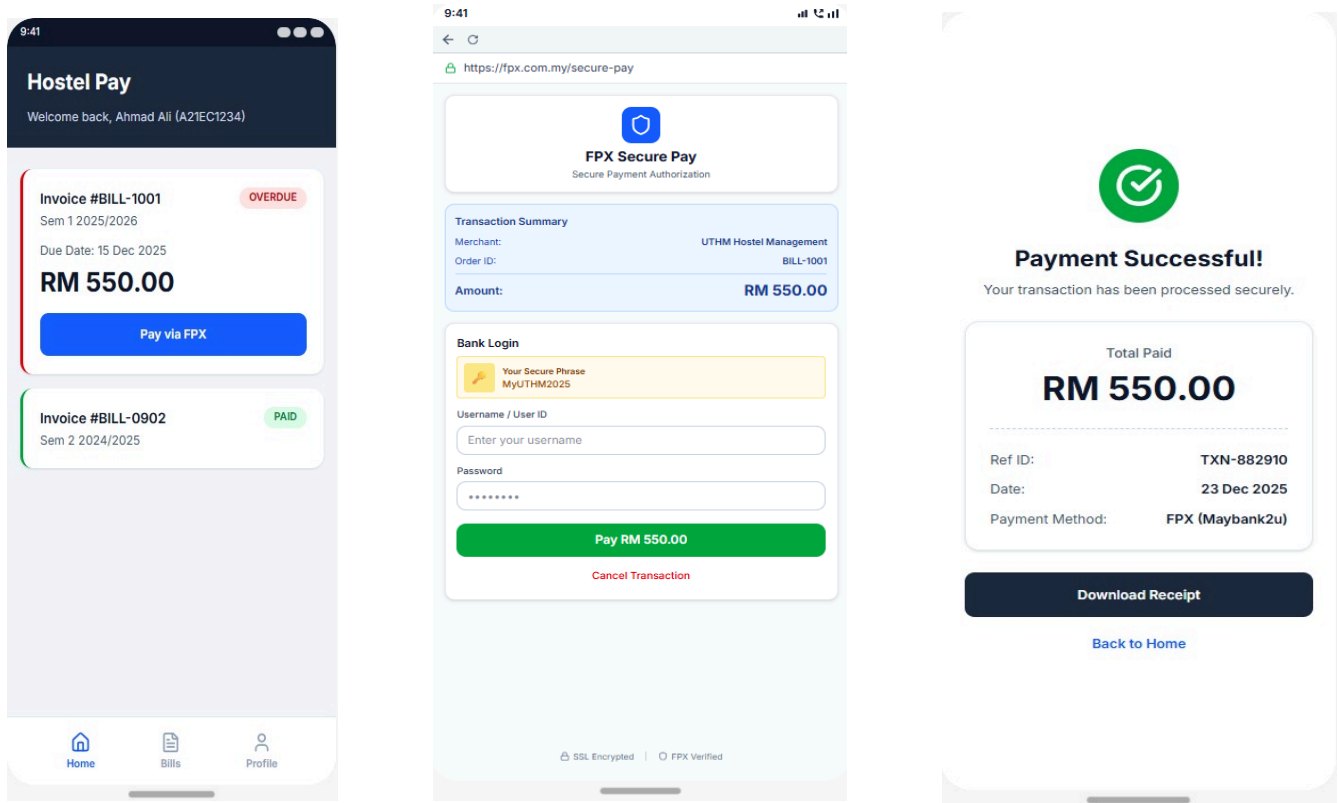
4.8.6.1 Class Diagram - Payment & Billing Tracker



4.8.6.2 Entity Relationship diagram - Payment & Billing Tracker



4.8.7 Interface Design - Payment & Billing Tracker



4.9 Summary

This chapter has provided and analysed a detailed analysis and design plan of the UTHM Enterprise Architecture project. It commenced with the evaluation of the shortcomings of the system and operational issues of the existing system, and then the functional and non-functional requirements were defined based on the UTHM industry talk. The chapter proceeded to describe the strategic change between the As-Is and the To-Be architecture, which is an SAP BTP cloud-based solution. Lastly, the Payment & Billing Tracker subsystem was described in detail with the use of the UML diagrams and database models to guarantee a scalable, secure, and user-centric digital solution.

6.1 Introduction

The chapter serves as a conclusion of the Enterprise Systems Design and Modeling project, which is a summary of the building of the University Enterprise Architecture (UEA) of University Tun Hussein Onn Malaysia (UTHM). The main aim of this paper was to examine the shortcomings of the current (As-Is) information architecture of the university and develop a robust (To-Be) (UEA) Enterprise Architecture design that would support the vision of UTHM in the year 2030.

The project analyzed the four most important areas of enterprise architecture which are Business, Data, Application, and Technology based on TOGAF. This helped to reveal the major operational problems such as the data redundancy, fragmentation of the systems and manual administrative processes. To solve these problems, hybrid architecture was proposed to be built on the SAP Business Technology Platform (BTP) to integrate these fragmented functions into a unified system.

The project ended with technical implementation of the Hostel Management System to approve the viability of this architectural design. There are 8 subsystems for the hostel management system. These subsystems show the prototype of how the integration of packages, central databases and responsive user interfaces can effectively solve the inefficiencies of older environments. The chapter will also give a summary of the system key contributions, evaluate the constraints that were faced during the development phase and give a strategic roadmap on how the proposed architecture can be fully implemented in future.

6.2 System Contribution/Achievement

The new enterprise architecture (EA) and its built in subsystems are going to make an impact on the digital transformation process of UTHM. The main successes and contributions are the following ones:

Digital Goal Strategic Alignment: The system is direct to Digital UTHM vision of the university, which seeks to be 82% digitalized by 2030. It also correlates IT infrastructure and the KPIs of the Ministry of Higher Education (MOHE), as well as it can make the university top in the field of modern digital systems.

Better Operational Performance: The architecture eliminates most manual and paper-based systems by integrating the disjointed systems into the rational work packages. This can save the time used by the administration and avoid any duplication of work among departments.

Enhanced End-User Experience: Login feature has been addressed by using the Unified Identity Provider (IdP) and Single Sign-On (SSO). Students and staff are able to receive various services through a single safe portal with high security.

Transition to Paperless and Cashless Campus: The UTHM system achieves dual benefits through its automatic enrollment process and digital payment record integration into Smart

Campus Services because it decreases the university's carbon emissions while updating student-vendor financial transactions.

Data Governance and Privacy Stronger: The To-Be architecture is putting in a formal data governance framework for PKU data access protocols and data API integration standards. This ensures all sensitive details are secured and handled according to global best practices.

Centralized and Managed IT Infrastructure: The shift from a client-server model to a centralized user access marks a major technical milestone which helps to optimize the university's network and adoption of the hybrid infrastructure to allow their ICT team to manage the entire digital ecosystem from a single point of control. This results in faster troubleshooting, more efficient software development, and a more stable digital environment.

6.3 System Constraint

According to the system architecture presented, the system will work within limitations that are stipulated by the fact that the system will be based on the SAP Business Technology Platform (BTP) ecosystem and will be low-code. The architecture is directly tied to the functionality of particular SAP implementations, SAP Build Apps to interface, SAP HANA Cloud to centralized storage and SAP Analytics Cloud to reporting- that is, any non-standard low-code customization could be restricted. Also, though the system itself is modeled as able to operate in real time with cloud-native scalability, the current implementation stage is based on a dummy data model to model high load cases (e.g. annual trends), which suggests that the overall analysis of historical data is at the moment limited by the lack of a filled live database. Lastly, cloud connectivity of the system to facilitate real-time synchronization between the student and warden apps presupposes the reliance on the availability of sound internet connection on all operational layers.

6.4 Future Suggestion

6.5 Individual Reflection

6.6 Summary

Chapter 6 concludes the Enterprise Systems Design and Modeling project by summarizing the development of the University Enterprise Architecture (UEA) for UTHM. The project successfully transitioned from an analysis of the disjointed As-Is architecture to a salad To-Be design based on the Digital UTHM 2030 vision. The proposed hybrid architecture will use the TOGAF framework and the SAP

Business Technology Platform (BTP) to overcome such important inefficiencies as data duplication and manual operation.

In terms of Strategic Impact, this project is in line with the Ministry of Higher Education (MOHE) KPIs where in a bid to create a paperless and cashless campus and enhance operational efficiency by creating unified systems, and Single Sign-On (SSO) features.

Secondly, regarding operation improvements, switching to a centralized IT Infrastructure and formal data governance framework, the university would have a more dominant control over its own digital ecosystem, which would provide privacy of the data and can reduce the time of troubleshooting.

Lastly, the implementation of the Hostel Management System and its eight subsystems is one of the successful prototypes. It can demonstrate that a central database and integrated packages can solve the shortcomings of a legacy system.

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